

Exercise 1 :

You are given the following four domains:

- i. A group of people playing poker.
- ii. A person driving a car.
- iii. A machine detecting chocolate bars weighing less than 50g.
- iv. A doctor performing a medical diagnosis.

Classify each of the domains above along with the domain properties that we discussed in the lecture by entering yes/no in the cells of the following table:

	Accessible	Deterministic	Episodic	Static	Discrete	Single agent
Poker						
Car						
Machine						
Doctor						

Exercise 2 :

For each of the following assertions, say whether it is true or false and give a short explanation in 1 or 2 sentences for each of your answers.

- (a) An agent that senses only partial information about the state cannot be perfectly rational.

- (b) There exist task environments in which no pure reflex agent can behave rationally.

- (c) There exists a task environment in which every agent is rational.

- (d) The input to an agent program is the same as the input to the agent function.

- (e) It is possible for a given agent to be perfectly rational in two distinct task environments.

- (f) Every agent is rational in an unobservable environment.

- (g) A perfectly rational poker-playing agent never loses.

- (h) Suppose an agent selects its next action uniformly at random from a set of possible actions. There exists a deterministic task environment in which this agent is rational.